Ishita Sinha

+918252732075 | GitHub | ishitasinha16@gmail.com | LinkedIn

AI Engineer with hands-on experience architecting and shipping agentic AI applications, including LLM-based inference pipelines, RAG workflows, and multi-agent systems using LangChain, LangGraph, CrewAI, and open-source LLMs served via Groq. Experienced in building and evaluating AI solutions within a healthcare/medical device enterprise environment at Boston Scientific, translating real stakeholder needs into production-ready systems. Skilled in prompt engineering, model evaluation, and REST API development, with a strong foundation in Python and a bias toward shipping iteratively and improving system reliability.

SKILLS

- **AI/ML & LLM Frameworks:** LangChain, LangGraph, CrewAI, Pydantic, Groq(open-source LLM inference), PyTorch, TensorFlow, Scikit-learn, Transformer Architecture, Prompt Engineering, Context Engineering, Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, Agentic AI Architecture, NLP, End-to-End GenAI System Design
- **Databases & Vector Stores:** SQL. NoSQL (MongoDB), MySQL, ChromaDB, Firebase
- **APIs & Backend Development:** REST APIs, FastAPI, JSON, NodeJS, ExpressJS
- **Machine Learning & Statistical Techniques:** Regression, Classification, Clustering, Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation
- **Data Analysis & Engineering:** Data Cleaning, Data Transformation, Exploratory Data Analysis (EDA), Data Modeling, Statistical Analysis, ETL Pipelines, KPI Analysis
- **Business Intelligence & Visualization:** Microsoft Power BI, Tableau, Microsoft Excel, Dashboard Development, Interactive Reporting
- **Languages & Tools:** Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn), SQL, Java, JavaScript, GitHub, Jupyter Notebook, VS Code, Postman, Figma, Selenium, Microsoft Power Automate

EXPERIENCE

Boston Scientific

Data Analyst/AI Engineer — *Aug 2024 – Present*

- Architected end-to-end LLM-based inference pipelines and agentic AI workflows using LangChain and LangGraph, reducing manual analysis time by ~70% and enabling 40+ stakeholders to surface actionable insights from unstructured enterprise data.
- Designed and deployed predictive ML models, including regression-based forecasting and supervised classification, to identify critical performance drivers and patterns informing decision-making across architecture and leadership teams.
- Built AI-ready data pipelines using Tableau Prep Builder, automating data cleaning, transformation, and feature engineering across 4+ data sources, cutting per-cycle processing time by ~70%.
- Implemented RAG-inspired contextual retrieval architectures integrating structured and unstructured data sources to enhance LLM response quality and enable domain-specific, context-aware inference at scale.
- Defined data collection, storage, and consumption strategies for next-generation AI use cases in collaboration with enterprise architecture teams, supporting reporting infrastructure used by 40+ stakeholders across the organization.

IT Intern — *Jan 2024 –Jun 2024*

- Developed interactive analytics dashboards integrated with enterprise collaboration tools, enabling leadership to make data-driven decisions; collaborated with Enterprise Architecture teams to define data collection, storage, and consumption strategies for reporting workflows.
- Engineered ETL processes using Excel, Power Query, and DAX to transform raw data from 4+ sources into consistent schemas, and automated workflows using Microsoft Power Apps and Power Automate to improve efficiency.
- Designed a centralized SharePoint hub using Figma to organize 5+ enterprise dashboards, improving data discoverability, accessibility, and overall user experience.

Elastic Run

Software Developer Intern — *May 2023 – Jul 2023*

- Developed a User Access Management System using the Frappe Framework to manage user permissions and access logs across enterprise systems, improving traceability and supporting secure, efficient system operations.
- Designed and implemented custom Doctypes and leveraged database structures to organize user data, enabling scalable, secure management of user roles and activity logs.

Ishita Sinha

+918252732075 | GitHub | ishitasinha16@gmail.com | LinkedIn

- Conducted automated unit testing using Python, achieving 94% test coverage, which enhanced application reliability and reduced potential system errors.

PROJECTS

ArchLens | GitHub

- Architected a generative AI architecture review platform using LangChain, LangGraph, and Groq-based LLM orchestration, scoring submitted architectures across 5 quantitative dimensions — architecture quality, security, scalability, cost efficiency, and operational readiness.
- Designed a 7-stage multi-agent LangGraph workflow sequencing an architecture analyzer, five specialist reviewer agents (security, cloud, data, DevOps, and enterprise architects), and a governance board stage to generate risk matrices and prioritized recommendations.
- Built a React frontend with centralized routing via React Router across 2 core views — an architecture intake portal and a review dashboard — keeping navigation logic separate from UI components.

PrepAI | GitHub

- Built an end-to-end interview preparation platform using LangChain and Groq LLMs, combining 4 AI-driven capabilities — roadmap generation, schedule planning, dynamic question generation, and answer evaluation — into a single workflow.
- Applied structured prompt engineering to classify questions across 3 difficulty tiers (Junior, Intermediate, Senior) and evaluate candidate responses across 3 dimensions — clarity, technical depth, and communication.
- Connected a React frontend to a FastAPI backend across 3 pages to handle the flow from candidate input to AI-generated feedback.

Behavioral Churn Radar Dashboard | GitHub

- Designed a two-page churn analytics dashboard in Power BI — Subscription Intelligence and Churn Risk Deep Dive — modeling a synthetic dataset of 50 simulated subscribers across 4 relational fact tables (subscriptions, user activity, support tickets, payments).
- Built 5 custom DAX measures and a calculated column — including Churn Rate %, Churn Risk Score, and Failed Payments — to power dynamic filtering, segmentation, and risk scoring.
- Performed churn, cohort, and behavioral analysis on the dataset, identifying a 24% overall churn rate (12 of 50 simulated subscribers) and surfacing engagement and payment-stability signals tied to churn risk.

CERTIFICATIONS

- **Preparing Data for Analysis with Microsoft Excel:** Learned data cleaning, structuring, and preparation techniques for analytical workflows.
- **Ask Questions to Make Data-Driven Decisions:** Learned structured problem-solving approaches for defining business questions, stakeholder requirements, and analytical objectives.
- **Using Python to Interact with Operating System:** Applied Python scripting to automate operating system tasks, manage files and processes, and work with system-level data. Utilized regular expressions, Bash scripting, and Python automation to streamline repetitive IT and system administration tasks.
- **Prepare Data for Exploration:** Gained hands-on experience in collecting, organizing, and structuring datasets using spreadsheets and database concepts. Applied techniques for data validation, data types management, and dataset preparation prior to analysis.
- **Generative AI with Large Language Models:** Gained in-depth understanding of the Generative AI lifecycle, transformer architecture, pre-training, fine-tuning (instruction tuning and RLHF), and deployment strategies for LLM-powered applications.
- **Prompt Engineering for ChatGPT:** Developed advanced prompt engineering techniques including few-shot prompting, chain-of-thought reasoning, and output structing to build reliable , task-specific LLM-powered workflows.
- **Building Systems with the ChatGPT API:** Learned to build multi-step LLM pipelines, chained reasoning systems, and evaluation frameworks using the ChatGPT API, with a focus on practical application design and safety best practices.

EDUCATION

- **Sikkim Manipal Institute of Technology** BTech (Computer Science) ***2024***